Unit: 6 Unit 7: Factoring
Topic: Sum and Difference of Cubes

Name: ____________
Date: __________

Foundation (Jalapeno)

Q1. What is the factored form of $x^3 + y^3$?
(A) $(x + y)(x^2 - xy + y^2)$
(B) $(x - y)(x^2 + xy + y^2)$
(C) $(x + y)(x^2 + xy - y^2)$
(D) $(x - y)(x^2 - xy - y^2)$
(E) $(x + y)(x^2 + y^2)$

Q2. What is the factored form of $8x^3 - 27y^3$?
(A) $(2x - 3y)(4x^2 + 6xy + 9y^2)$
(B) $(2x + 3y)(4x^2 - 6xy + 9y^2)$
(C) $(2x - 3y)(4x^2 - 6xy - 9y^2)$
(D) $(2x + 3y)(4x^2 + 6xy - 9y^2)$
(E) $(2x - 3y)(4x^2 + 9y^2)$

Q3. What is the factored form of $x^3 - 8y^3$?
(A) $(x - 2y)(x^2 + 2xy + 4y^2)$
(B) $(x + 2y)(x^2 - 2xy + 4y^2)$
(C) $(x - 2y)(x^2 - 2xy - 4y^2)$
(D) $(x + 2y)(x^2 + 2xy - 4y^2)$
(E) $(x - 2y)(x^2 + 4y^2)$

Q4. What is $a^3 + b^3$ equal to?
(A) $(a + b)(a^2 - ab + b^2)$
(B) $(a - b)(a^2 + ab + b^2)$
(C) $(a + b)(a^2 + ab - b^2)$
(D) $(a - b)(a^2 - ab - b^2)$
(E) $(a + b)(a^2 + b^2)$

Q5. What is $27x^3 + 8y^3$ equal to?
(A) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
(B) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
(C) $(3x + 2y)(9x^2 + 6xy - 4y^2)$
(D) $(3x - 2y)(9x^2 - 6xy - 4y^2)$
(E) $(3x + 2y)(9x^2 + 4y^2)$

Q6. Which of the following is a difference of cubes?
(A) $x^3 + y^3$
(B) $x^3 - y^3$
(C) $x^2 + y^2$
(D) $x^2 - y^2$
(E) $x + y$

Q7. What is the factored form of $64x^3 - y^3$?
(A) $(4x - y)(16x^2 + 4xy + y^2)$
(B) $(4x + y)(16x^2 - 4xy + y^2)$
(C) $(4x - y)(16x^2 - 4xy - y^2)$
(D) $(4x + y)(16x^2 + 4xy - y^2)$
(E) $(4x - y)(16x^2 + y^2)$

Q8. What is $8x^3 + 27y^3$ equal to?
(A) $(2x + 3y)(4x^2 - 6xy + 9y^2)$
(B) $(2x - 3y)(4x^2 + 6xy + 9y^2)$
(C) $(2x + 3y)(4x^2 + 6xy - 9y^2)$
(D) $(2x - 3y)(4x^2 - 6xy - 9y^2)$
(E) $(2x + 3y)(4x^2 + 9y^2)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^3 - 64y^3$

Q10. Prove the identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Chef's Tips

- Read the questions carefully
- Use the correct formulas
- Check your work